

Governed AI in a QC lab — a conversation guide

A one-page version of the interactive demo: how to talk a lab through AI assistance without giving up control of GxP-critical decisions. Fictional QC scenario (batch B-2047, HPLC, on hold) — not Thermo Fisher software.

The principle: the model proposes, the code decides.

The AI can *suggest* an action. A permission policy enforced in code — not in prompts — decides what runs automatically, what waits for a human, and what is never automated. Even a highly confident model never triggers an irreversible side effect on its own.

THE CUSTOMER SCENARIO, IN THREE SCENES

AUTO · SPEED

"Summarize the QC results."

Read-only and low risk. The AI answers instantly — no queue. The scientist gets speed.

NEVER · CONTROL

"Can the AI release my batch?"

Model is 92% confident — policy still says NEVER. Batch release is human-only, fail-closed. QA keeps control.

ASK · OVERSIGHT

"Flag this off-spec sample."

This writes to the quality record, so it is queued for QA review. Supervised, not blocked.

THE PERMISSION POLICY

ACTION	DECISION
Summarize results · query LIMS	AUTO
Flag anomaly · write ELN entry	ASK
Release batch · change LIMS status	NEVER
Unknown / unexpected action	NEVER

Fail-closed by default. Auto actions also require model confidence ≥ 0.70 , or they escalate to a human.

ONE WORKFLOW, THREE STAKEHOLDERS

STAKEHOLDER	WHAT THEY GET
Scientist	Speed — self-serve read-only answers, no friction.
QA & Compliance	Control — human sign-off on every write and every GxP-critical action.
IT	Safe integration — closed action set, policy in code, fail-closed.

The compliance story. Every step — received, proposed, gated, executed or queued — lands in an append-only log, and each row is hashed to the one before it. Edit history after the fact and the chain breaks on verification. You can prove what the system did and detect tampering.

Fictional QC scenario — not Thermo Fisher software. Same architecture used in production regulated clinical AI: closed action set, fail-closed permission policy, hash-chained audit. Interactive demo: ainoafv.github.io/regulated-lab-agent-demo

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Governed AI for regulated lab workflows